Do Harsher DUI Punishments Deter All Drivers Equally?

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1. Research Question

Hansen (2015) finds that crossing the 0.08 BAC threshold and receiving a DUI reduces future DUI recidivism.

Does this deterrent effect differ across drivers?

We extend the original regression-discontinuity design using an honest causal forest to study heterogeneity.

2. Regression-Discontinuity Design at 0.08 BAC

Washington's DUI law uses a 0.08 BAC threshold.

Drivers just below and just above 0.08 have nearly identical BAC levels, but those above 0.08 receive a DUI and face more severe sanctions.

Treatment indicator: $D_i = 1(BAC_i \geq 0.08)$.

We compare drivers on either side of the threshold, holding BAC constant, to estimate the causal effect of receiving a DUI.

$$Y_i = \alpha + \tau D_i + \beta_1(BAC_i - 0.08) + \beta_2 D_i(BAC_i - 0.08) + X_i'\gamma + \epsilon_i.$$

A negative τ means receiving a DUI reduces future recidivism.

3. Hansen's Original Result (0.08 Threshold)

−2.1 percentage points
Reduction in four-year DUI recidivism

Method	Estimate	S.E.
Hansen RDD	−0.021	0.004

- Statistically significant at the 1% level.
- Baseline recidivism near the threshold was about 10.3%.
- Implies roughly a 17% relative reduction.

4. How Our Analysis Extends Hansen

Aspect	Hansen (2015)	Our Extension
Main result	Overall local RDD effect	Overall effect + CATEs
Heterogeneity	A few predefined subgroup comparisons	Flexible, data-driven heterogeneity (CATEs)
Main subgroups	By prior DUI-test history (no tests vs. ≥1 test)	Age, sex, race, accident involvement, and year
Goal	Estimate average deterrence	Learn who benefits most

Our extension searches more flexibly for where the deterrent effect is stronger or weaker.

5. Key Data Features

- Modified Washington State DUI administrative data, 1999–2007.
- ML treatment examined: crossing the 0.08 DUI threshold.
- Hansen also studies the 0.15 aggravated-DUI threshold; not used in our main forest analysis.
- Outcome: DUI recidivism within four years.
- Main bandwidth: 0.05 around 0.08.
- Effect modifiers in the forest: age, sex, race, accident involvement, year.

Balance checks support the RDD: predetermined characteristics do not jump at the 0.08 threshold.

6. Honest Causal Forest (How It Works)

We estimate Conditional Average Treatment Effects (CATEs):

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x,\ BAC_i \approx 0.08].$$

Procedure:

Local sample around BAC = 0.08
↓
Remove local BAC trends
↓
Fit honest causal forest
↓
Estimate ATEs and CATEs

7. Average Effects Are Almost Identical

Method	Effect	95% CI
Hansen RDD	−0.0210	approx. [−0.029, −0.013]
Our RDD	**−0.0222**	**[−0.0303, −0.0140]**
Causal forest (ATE)	**−0.0226**	**[−0.0923, 0.0471]**

$$-0.0210 \approx -0.0222 \approx -0.0226$$

Both methods estimate an average reduction of about **2.2 percentage points.**

8. Are the Average Effects Meaningful?

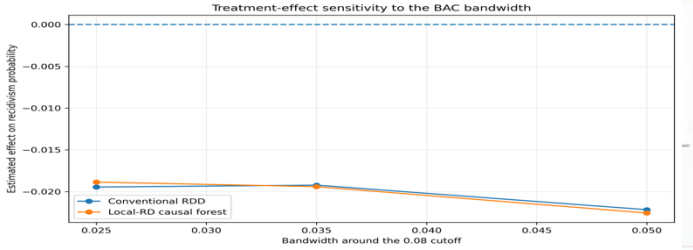
Statistical significance:
The RDD confidence interval excludes zero, so our replication estimate is statistically significant. The forest point estimate is similar but less precise (its interval includes zero).

Economic significance:
A 2.2 percentage-point decline from a 10–12% baseline rate implies:

18%–20% relative reduction.

This is large enough to matter for policy.

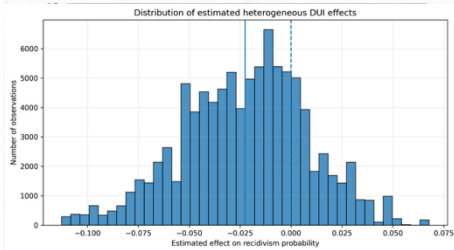
9. Robustness Across Bandwidths



10. Average-Effect Conclusion

Our ML estimator confirms Hansen's average deterrence result.
The value of the causal forest is the ability to study heterogeneity, not to replace the RDD estimate.

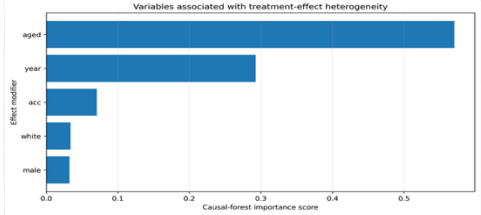
11. Estimated Heterogeneous Effects (CATEs)



Statistic	Estimate
Mean CATE	−0.0226
Median CATE	−0.0206
5th percentile	−0.076
95th percentile	0.028

More negative values imply stronger deterrence.
The CATE distribution suggests effects vary across drivers.

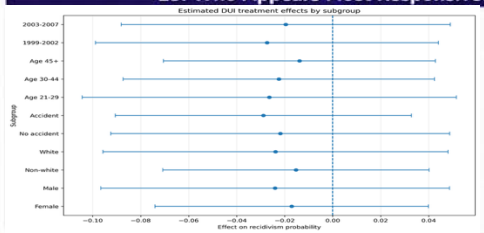
12. Which Characteristics Matter Most?



- **Age** is the most important effect modifier.
- **Year** is the second-most important modifier.
- **Sex**, race, and accident status matter less in the forest.

Variable importance shows how useful each characteristic was for splitting the forest into groups with different predicted treatment effects.
It is not a test of statistical significance or causality.

13. Who Appears Most Responsive?



Younger drivers show the strongest estimated deterrent effect.

Age 21–29: −0.026

Age 45+: −0.014

14. How Strong Is the Heterogeneity Evidence?

- Estimated effects vary most clearly by age.
- The pattern could matter for policy targeting.
- However, subgroup confidence intervals overlap.

These results are exploratory rather than definitive.

15. Main Conclusion

Punishment reduces recidivism on average.

The causal forest reproduces Hansen's ATE and suggests the effect may be stronger for younger drivers.

The average effect is statistically and economically meaningful; the heterogeneity evidence is suggestive but less definitive.

Main result: Our RDD and causal forest both estimate an average reduction in DUI recidivism of about 2.2 percentage points.

Hansen (2015), American Economic Review